

ECEN-662: Homework 8

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Problem 1

Given a uniform random variable $U \sim \mathcal{U}([0, 1])$, derive the inverse transform sampling function required to generate samples from a Rayleigh distribution with parameter σ :

$$f(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad x \geq 0.$$

Solution

Calculating CDF (with help from WolframAlpha):

$$\begin{aligned} F(x) &= \int_0^x \frac{t}{\sigma^2} \exp\left(-\frac{t^2}{2\sigma^2}\right) dt \\ &= 1 - \exp\left(-\frac{x^2}{2\sigma^2}\right) \end{aligned}$$

Now equating with U :

$$\begin{aligned} U &= F(x) = 1 - \exp\left(-\frac{x^2}{2\sigma^2}\right) \\ U - 1 &= -\exp\left(-\frac{x^2}{2\sigma^2}\right) \\ 1 - U &= \exp\left(-\frac{x^2}{2\sigma^2}\right) \\ \ln(1 - U) &= -\frac{x^2}{2\sigma^2} \\ x &= \sigma \sqrt{-2 \ln(1 - U)} \end{aligned}$$

Problem 2

Suppose we want to sample from a standard Normal distribution, but only the positive half (a truncated normal). The target density

$$f(x) \propto \exp(-x^2/2), \quad x > 0.$$

We will use an exponential distribution

$$g(x) = \lambda \exp(-\lambda x), \quad x \geq 0$$

as our proposal distribution.

- (i) Find the optimal scaling constant M as a function of λ such that $f(x) \leq M g(x)$ for all $x \geq 0$.
- (ii) Analytically determine the optimal rate parameter λ that minimizes the rejection rate.

Solution

Part (i): Setting the ratio $h(x) = \frac{f(x)}{g(x)}$:

$$h(x) = \frac{\exp(-\frac{x^2}{2})}{\lambda \exp(-\lambda x)} = \frac{1}{\lambda} \exp\left(-\frac{x^2}{2} + \lambda x\right).$$

And taking the derivative and equating to zero:

$$\frac{d}{dx} \left(-\frac{x^2}{2} + \lambda x \right) = -x + \lambda = 0 \Rightarrow x = \lambda.$$

Now plugging back in:

$$M(\lambda) = \frac{1}{\lambda} \exp\left(\frac{\lambda^2}{2}\right)$$

Part (ii): Differentiating the logarithm of $M(\lambda)$ and equating to zero:

$$\frac{d}{d\lambda} \ln M(\lambda) = \frac{1}{M(\lambda)} \frac{d}{d\lambda} M(\lambda) = -\frac{1}{\lambda} + \frac{\lambda}{2} = 0$$

Solving for λ :

$$\lambda^2 = 2 \Rightarrow \lambda = \sqrt{2}$$

Problem 3

We want to estimate the probability of a standard normal random variable exceeding a threshold of 4:

$$\mathbb{P}(X > 4) = \int_4^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

- (i) If we use standard Monte Carlo by drawing $N = 10^5$ samples from $X \sim \mathcal{N}(0, 1)$, what is the expected number of samples that will fall in the region $X > 4$?
- (ii) Propose a shifted Gaussian importance sampling distribution

$$q(x) = \mathcal{N}(\mu, \sigma^2)$$

that would significantly reduce the variance of the estimate. Justify your choice of μ and write out the final Importance Sampling estimator equation.

Solution

Part (i): The expected number of samples is given by

$$\begin{aligned} \mathbb{E}[X > 4] &= n\mathbb{P}(X > 4) = 10^5 \int_4^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx \\ &= 3.167 \end{aligned}$$

Part (ii): A Gaussian distribution with mean $\mu = 4$ and variance $\sigma^2 = 1$ would be an improvement as it is centered around the target region.

Problem 4

Let the target distribution be an unnormalized symmetric bimodal distribution:

$$\pi(x) \propto \exp\left(-\frac{(x-3)^2}{2}\right) + \exp\left(-\frac{(x+3)^2}{2}\right).$$

We will use a Metropolis-Hastings algorithm with a Gaussian random walk proposal:

$$p(x, x') = \mathcal{N}(x'; x, \sigma_p^2).$$

Assume the current state of the chain is $x_t = 3.0$ (sitting precisely on the right peak).

- (i) Calculate the exact acceptance ratio $\alpha(x_t, x')$ if the proposal draws a new state $x' = 0.0$ (the deep valley between the peaks).
- (ii) Conceptually, what happens to the Markov Chain if the proposal variance σ_p^2 is set extremely small (e.g., 0.01)? What happens if it is set extremely large (e.g., 100)?

Solution

Part (i): The acceptance ratio is given by

$$\begin{aligned} \alpha(x_t, x') &= \min\left(1, \frac{\pi(x')p(x', x_t)}{\pi(x_t)p(x_t, x')}\right) \\ &= \min\left(1, \frac{\exp(-\frac{(0-3)^2}{2}) + \exp(-\frac{(0+3)^2}{2})}{\exp(-\frac{(3-3)^2}{2}) + \exp(-\frac{(3+3)^2}{2})}\right) \\ &= \min\left(1, \frac{2 \exp(-4.5)}{1 + \exp(-18)}\right) \approx 0.022 \end{aligned}$$

Part (ii): If the proposal variance is set extremely small, the Markov Chain will make only small moves, lowering exploration. With extremely large proposal variance, many proposed states will result in rejection, reducing exploration.